UC San Diego
JACOBS SCHOOL OF ENGINEERING
Mechanical and Aerospace Engineering

District Heating Cooling Systems

MAE 119, Spring 2025

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Introduction to "Decarbonized district energy systems: Past review and future projections" by Simpson, Juliet G., et al.

1 General Overview of DES

- **District Energy Systems (DES)** utilize "...centralized boilers for hot water or steam and electric-based chillers for chilled water…"
- **Thermal Energy** is distributed to buildings within a district using **shared infrastructure** and resources
- DHC historically produces substantial carbon emissions, but technological advancements have allowed for **reduced emissions** or **completely decarbonized** DES to be possible

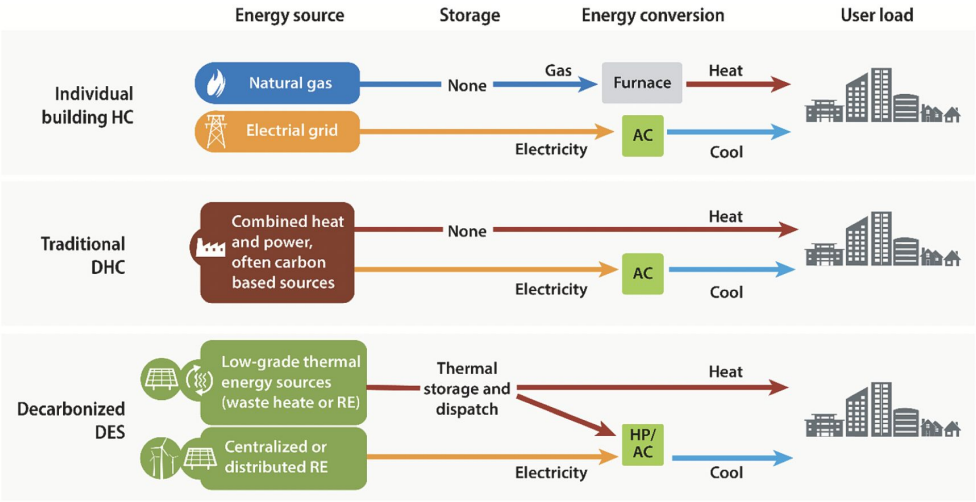

Figure: Overview of heating and cooling options for individual or district systems

Simpson, Juliet G., et al. "Decarbonized District Energy Systems: Past Review and Future Projections." Science Direct, Elsevier, 24 Sept. 2024, www.sciencedirect.com/science/article/pii/S2590174524002046.

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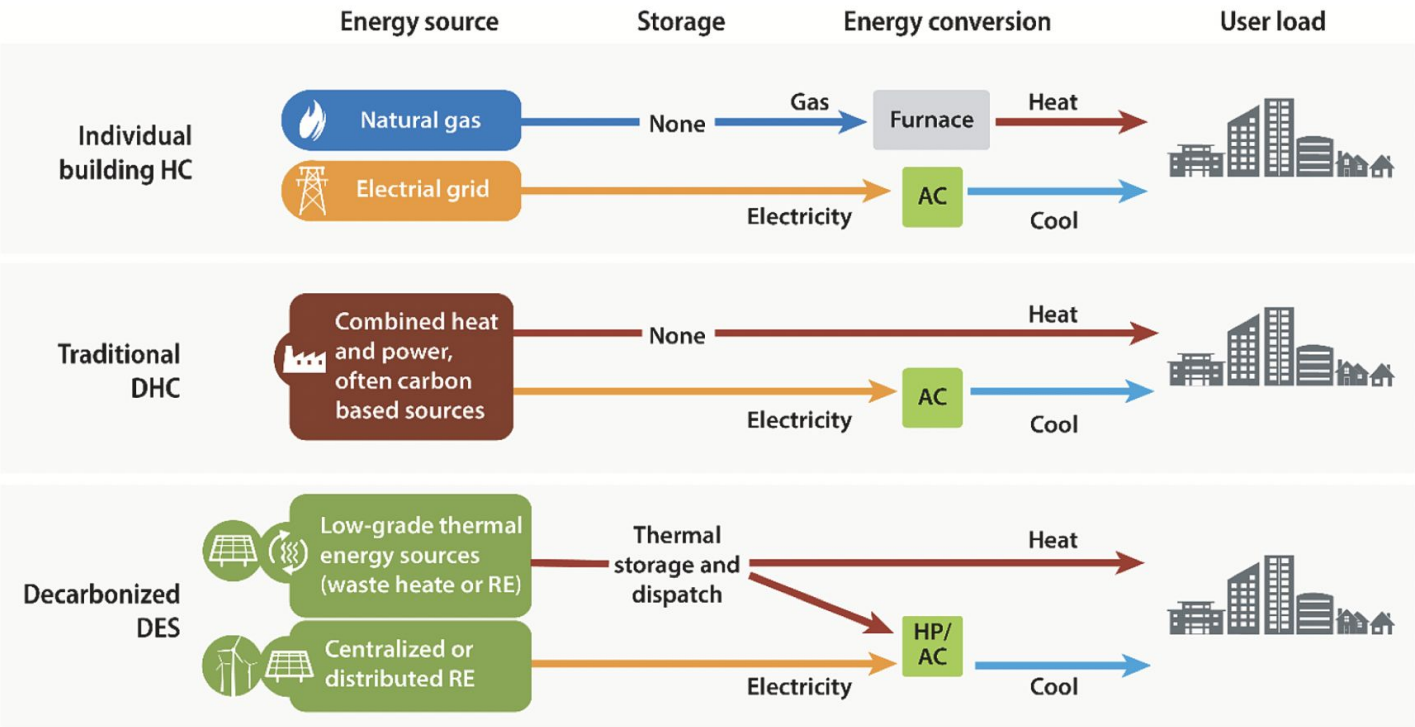

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1 Different forms of DES

- A main difference between DES (1G-5G) is their operating temperatures. **With lower temperatures systems become much more efficient.**
- Due to their lower temperatures 4G and 5G systems can take in waste heat from sewage, data centers etc...
- 5G systems use end-user heat pumps to increase efficiency due to that their coefficient of performance (COP) > 1.
- Radial, ring, and meshed grid types escalate in their complexity offering better energy optimization and scalability with mesh supporting 5G best

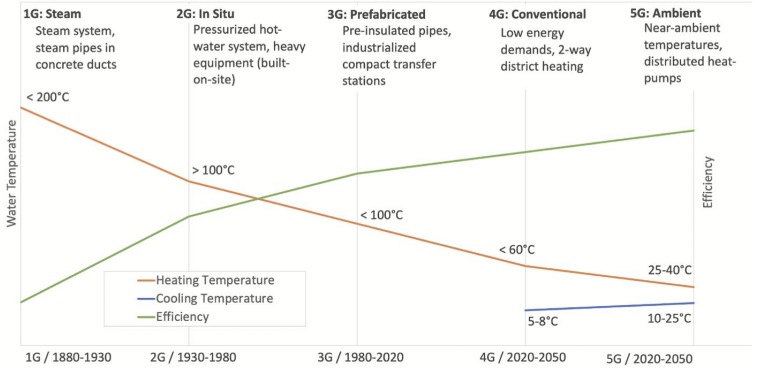
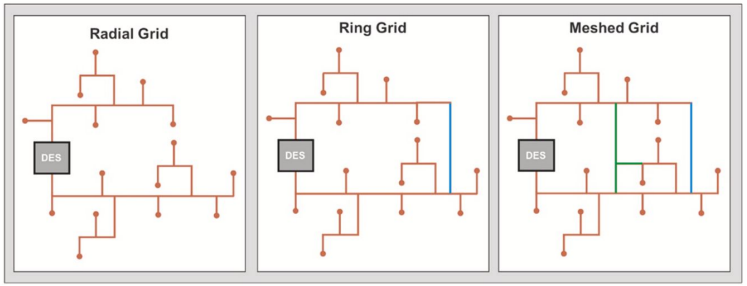


Fig. 2. Temperature and efficiency of various generations of district energy systems (adapted from [15])



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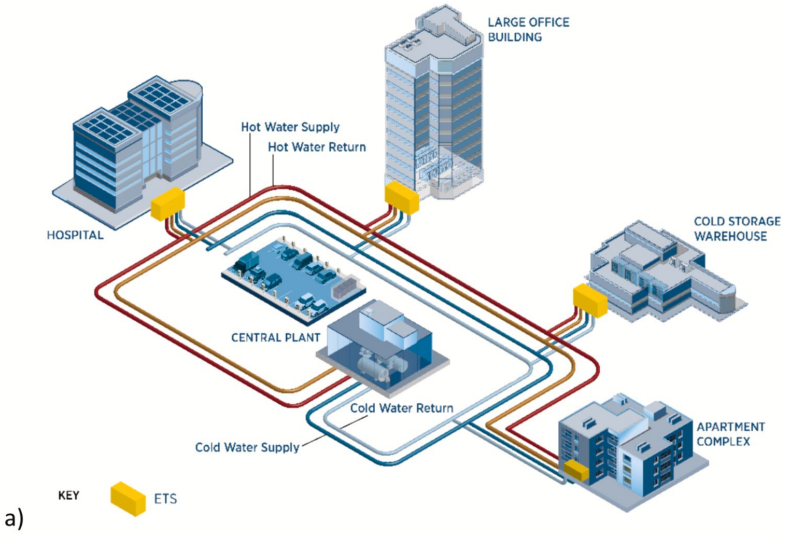
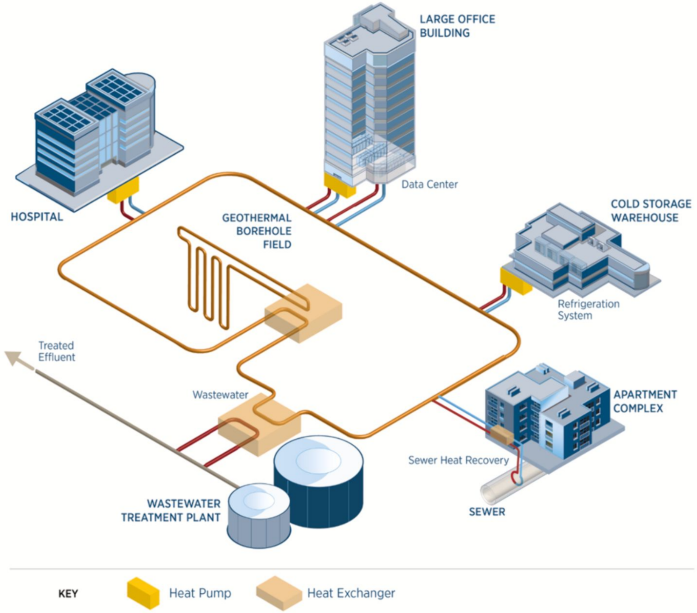
1 4G vs 5G systems

4G DES delivering hot and cold water to buildings via **separate return and supply pipes.** Energy transfer station (ETS) converts water supply to space heating/cooling



a)

Bi-directional, one pipe, 5G DES integrating geothermal storage, waste heat capture, and water source heat pumps to increase temperature.



b)

Simpson, Juliet G., et al. “Decarbonized District Energy Systems: Past Review and Future Projections.” Science Direct, Elsevier, 24 Sept. 2024, www.sciencedirect.com/science/article/pii/S2590174524002046.

Technical Description of Contributions

2 Sources & Storage

- **Continuous sources** such as biomass or coal provide stable energy to the system but are mainly based in fossil fuels or nuclear energy.
- **Variable sources** such as solar and wind are tougher to integrate stably however, they can offer benefits when combined with DES and thermal storage to facilitate effective demand management.
- **Other sources** include waste heat, solar thermal energy, and natural reservoirs like lakes.
- Storage can play an important role in DES by providing a **buffer for demand and variability of electricity sources.**
  - Short term **"thermal inertia"** storage is contained within the system loop through widening temperature setpoints.
- **Geothermal** is a key focus for its application in higher temperature systems, heat exchangers, and thermal energy storage.

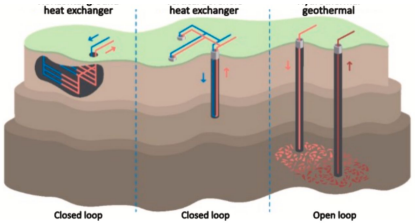


Fig. 5. Common types of geothermal systems used for district heating and cooling: shallow horizontal or vertical heat exchanger and deep hydrothermal. Adapted from [38].

Simpson, Juliet G., et al. "Decarbonized District Energy Systems: Past Review and Future Projections." Science Direct, Elsevier, 24 Sept. 2024, www.sciencedirect.com/science/article/pii/S2590174524002046.

2 Real World Systems

Location and year installed	System type	Key sources and storage	Thermal conversions
Ball State University. Indiana, USA 2012	3 Pipe 3G	3,600 boreholes at 122 m deep. Centralized heat recovery chillers	heat exchangers at users
Drake Landing Solar Community. Alberta, CA	4G DES, two pipes	144 boreholes at 35 m deep. 240 m3 hot water storage. Distributed solar thermal collectors	heat exchangers at users

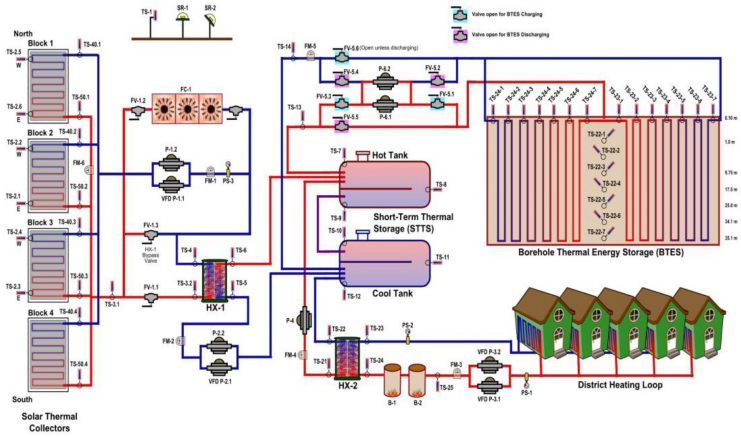


Fig. 2. Functional system schematic with monitoring points

Drake Landing Solar Community schematic

Information presented is an abbreviated version of Table 1 in Simpson, Juliet G., et al.

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2 Real World Systems

Location and Year installed	System type	Key sources and storage	Thermal Conversions
Whisper Valley "GeoGrid." Texas, USA. 2017	5G DES, ambient temperature	37 boreholes at 107 m deep distributed along loop. Cooling tower	heat pumps at users
Colorado Mesa University. Colorado, USA 2008	5G DES, one pipe loop, ambient temp	471 boreholes, 91–152 m deep. Cooling towers, irrigation HX	heat pumps at users

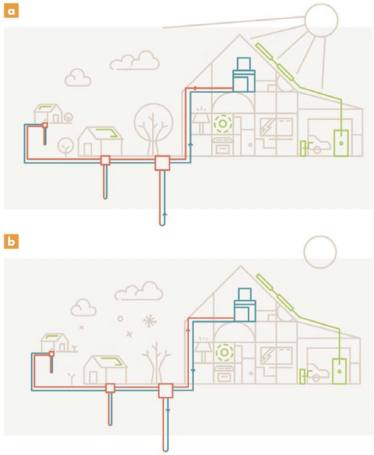


Figure 3 (Click to View Larger): The EcoSmart GeoGrid district-wide geothermal exchange system links hundreds of individual ground loops to create a highly efficient, integrated thermal management system during summer (a) and winter (b).

Whisper Valley system by season

CMU circulation and heat pumps

(b) (c)

Figure 3. Central geothermal pump station at CMU's Wubben Science Hall: (a) central remote monitoring system, (b) three circulation pumps, and (c) heat pump

Information presented is an abbreviated version of Table 1 in Simpson, Juliet G., et al.

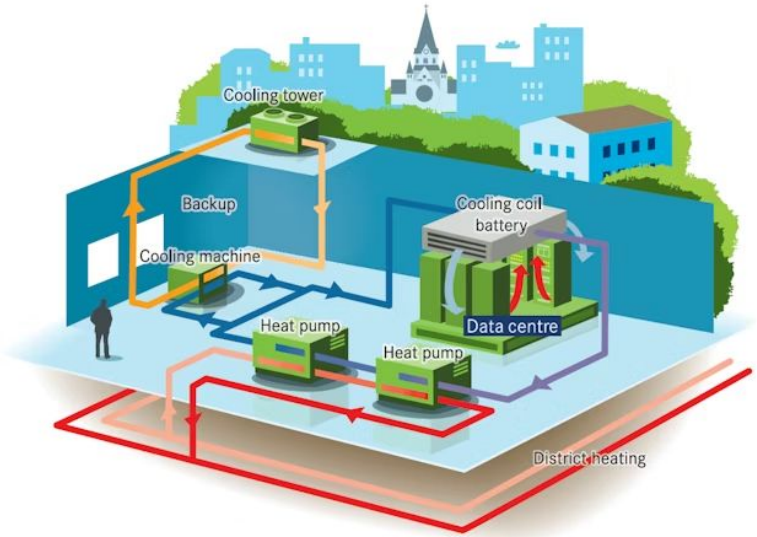
The Importance of Nanogrids in Low-Carbon Residential Communities - WhisperValleyAustin.com. (2020, September 29). Whisper Valley Austin. https://www.whispervalleyaustin.com/the-importance-of-nanogrids-in-low-carbon-residential-communities/

Oh, H., & Beckers, K. (2023). *Cost and Performance Analysis for Five Existing Geothermal Heat Pump-Based District Energy Systems in the United States*. https://doi.org/10.2172/1992646

Advantages and Limitations

3 Advantages

1. Reduces Greenhouse Gas Emissions
2. Supports Renewable Energy Integration and Grid Flexibility
3. Enables Use of Waste Heat and Low-Grade Thermal Sources



Heat pumps in district heating and cooling systems – Analysis - IEA. (2020). IEA. https://www.iea.org/articles/heat-pumps-in-district-heating-and-cooling-systems

3 Reduction in Greenhouse Gas Emissions

- Stanford **decreased** GHG emissions by **65%** with a 5th generation DHC installation.
- Colorado Mesa University and Ball State University **decreased** GHG emissions by **75%** and **19%**.
- Widespread use of geothermal heat pumps enables electrified heating and cooling while **reducing grid costs**, **transmission needs**, and **emissions**.

John Anderson. (2015, April 27). New heat-recovery system makes Stanford one of world's most energy-efficient uni's. New Atlas. https://newatlas.com/stanford-sesi-heat-capture-system/37184/

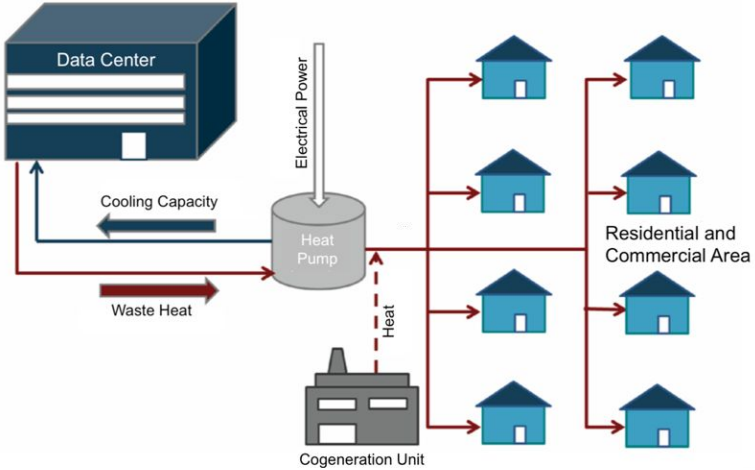
3 Grid Flexibility With Renewable Integration

- In Esbjerg, Denmark, large-scale **wind-powered** heat pumps are being used to charge **thermal energy storage**, helping to balance the grid.
- Electrically powered **heat pumps** can be strategically run during periods of **low-cost** or **surplus electricity**, especially from renewables.
- Thermal storage can **absorb excess renewable energy**, then discharge it later to meet heating and cooling demands reducing grid strain.

(2025). En-Former.com. https://www.en-former.com/en/danish-esbjerg-puts-trust-in-one-of-worlds-largest-heat-pumps/

3 Enables Use of Waste Heat and Low-Grade Thermal Sources

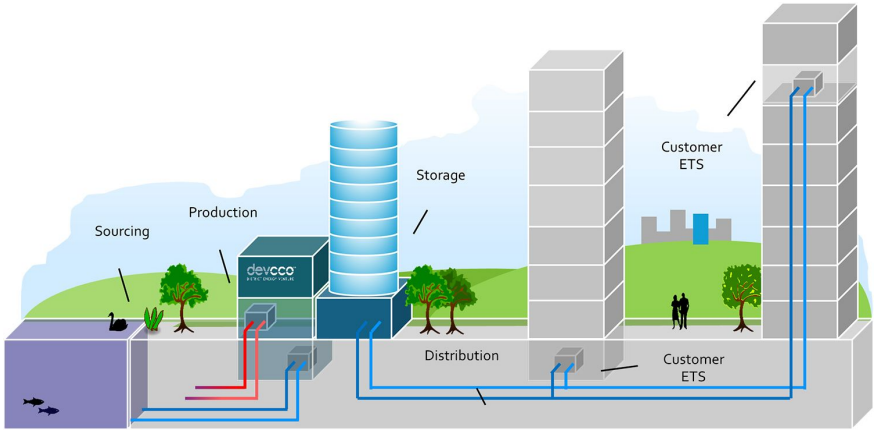
- “Unlike high-temperature systems, 4G and 5G systems operate at **lower temperatures** and can thus utilize waste heat from a large variety of sources.”
- Waste heat from **data centers**, **refrigeration**, and **sewage** can be integrated into the DHC loop.
- Naturally occurring heat sources such as **geothermal** can aid **heating** and lakes, rivers, and oceans can provide **cooling**.



Heat Recovery from Data Centers: A Win-Win Situation. (n.d.). Home. https://www.dotmagazine.online/issues/on-the-edge-building-the-foundations-for-the-future/heat-recovery-from-data-centers

3 Limitations

1. High Capital Costs
2. Complex Engineering and Site-Specific Design
3. Building Compatibility and Retrofitting Challenges



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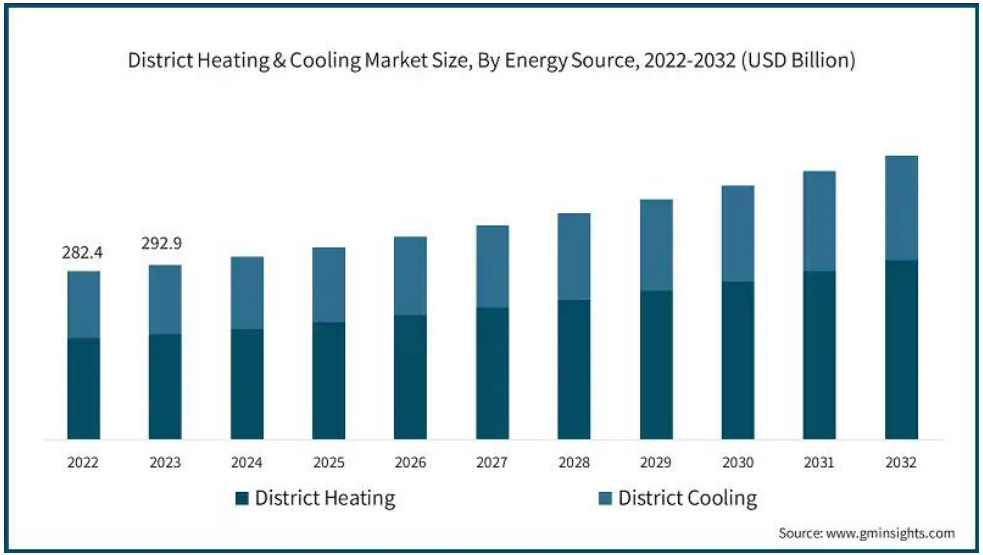
3 High Capital Costs

- "The **high capital cost** of switching to heat pumps at each end user makes the LCOE of 5G systems **prohibitively high** in many cases."
- DHC systems require very **high initial investment** in infrastructure such as distribution networks and centralized systems.



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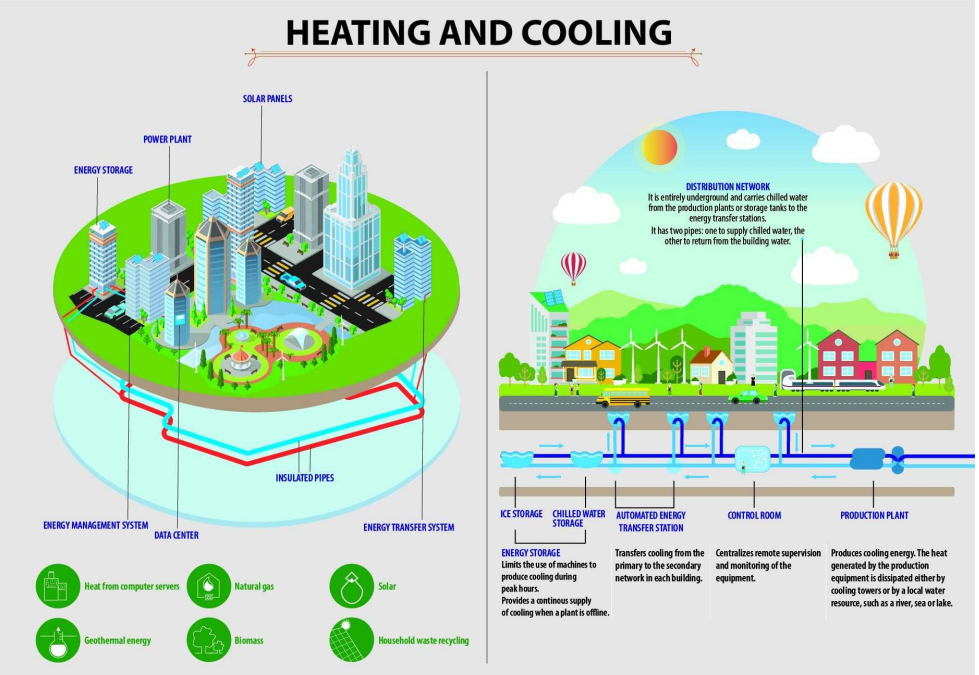
3 Site-Specific Engineering Challenges

- “Currently, DES designs are unique for each installation and **require tailoring** for each site.”
- Each DHC system must be **uniquely design** for its site. There is no one size fits all approach.
- Local conditions such as **climate** and **geography** heavily impact the sites design.
- **Highly skilled** engineering teams are required.



District Heating and Cooling System | Tek-Trol Technology Solutions. (n.d.). Tek-Trol. https://tek-trol.com/markets/district-heating-and-cooling/

3 Building Compatibility and Retrofitting Challenges

- Many existing buildings may **not** be **compatible** with DHC loop temperatures.
- **Retrofitting** to add heat-pumps can be very costly and in some situations **impossible** in **dense** or **historic areas**.
- "5G DES require **larger pipes** and **pumps**... and may incur higher capital cost to switch over the building systems to heat pumps."

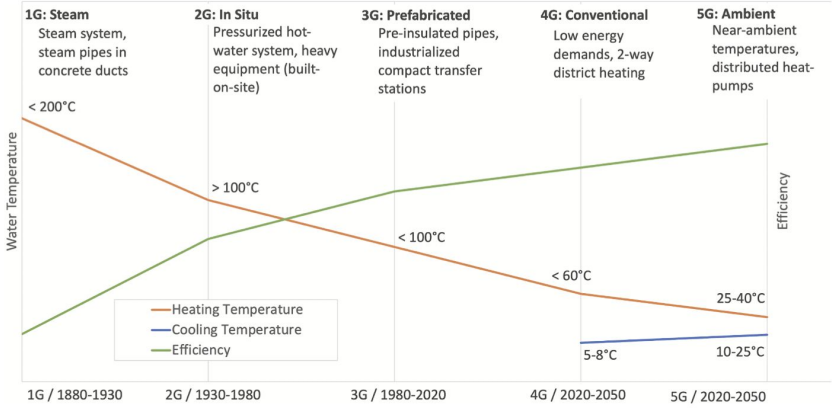


Fig. 2. Temperature and efficiency of various generations of district energy systems (adapted from [15])

Lalor, J., & Gillich, A. (2024). Retrofitting a Fifth Generation District Heating and Cooling Network for Heating and Cooling in a UK Hospital Campus. Buildings, 14(8), 2442. https://doi.org/10.3390/buildings14082442

Relevance of Contributions

4 Relevance of Contributions

- Integration of DES with the latest technology to achieve decarbonization
  - Renewable Thermal energy sources like geothermal and solar thermal
  - Waste heat recovery systems and heat pumps
  - Smart grid integration for controls, improved efficiency of system, and proper distribution of components
- Economic projections of market behavior as these systems come to be more in demand

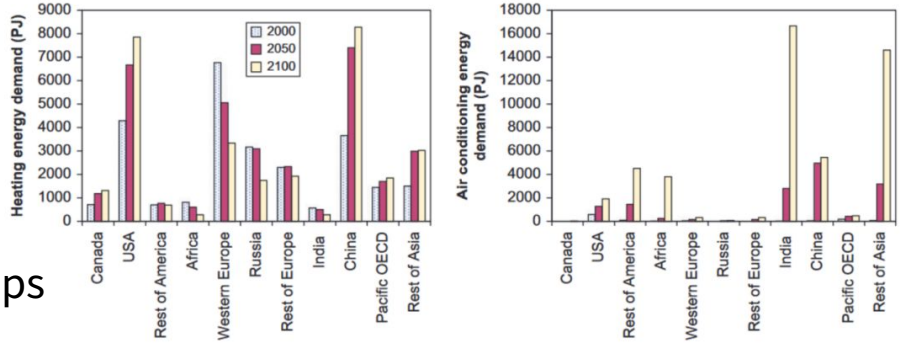


Figure: Projected heating and cooling demands in next 100 years, with heating demand decreasing in Northern regions and cooling demand increasing in developing regions

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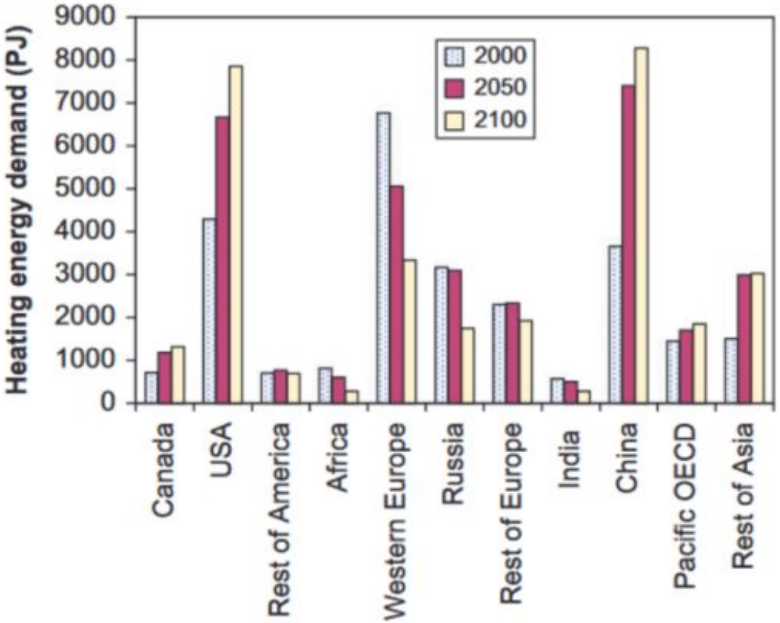

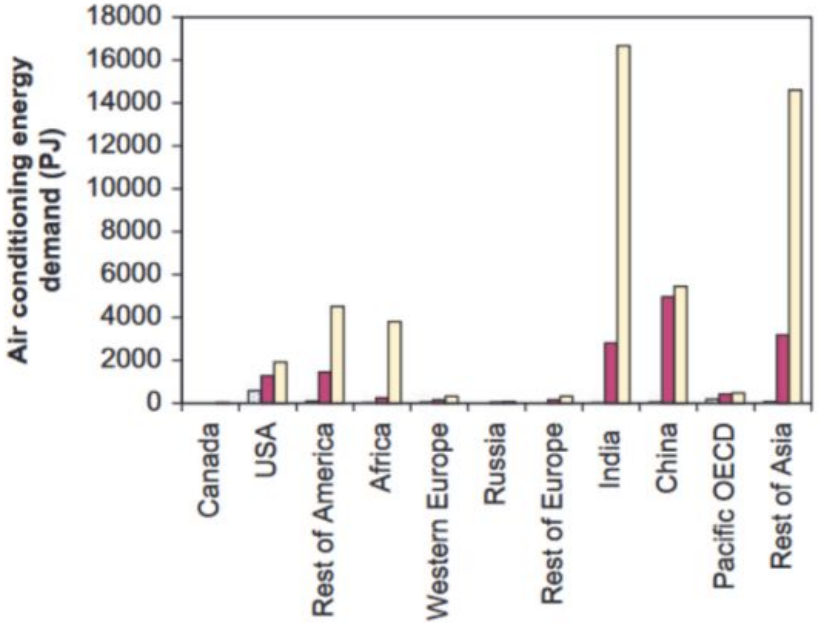

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4 Impact on the field

- District Energy Systems are less common in US, existing sites are mostly 4 or 5G installed in last 25 years, trends towards seeking decarbonization
- Research facilities given framework for improvements to their systems
- UCSD Cogeneration power plant
  - "Waste" heat and electricity used to generate steam and chilled water for some campus heating and cooling, a form of District heating and cooling, which could be decarbonized to create a DES system

Figure: UCSD Central Utilities Plant

Matching Load and Generation at UCSD. (2010, November). POWER Magazine. https://www.powermag.com/matching-load-and-generation-at-ucsd/

4 Real-World Implications

- Paper published by researchers at the National Renewable Energy Laboratory (NREL)
- NREL can…
  - Expand use of **URBANopt**, a simulation software designed to assist in planning DHC in the US
  - Support individual municipalities seeking to implement this specialized technology
  - Provide policy recommendations to local, state, and national governing bodies

URBANopt Advanced Analytics Platform | Buildings | NREL. (2025). Nrel.gov. https://www.nrel.gov/buildings/urbanopt

Unanswered Questions & Extensions of Research

5 Unanswered Questions

- Current DES designs are **highly customized** to each installation site.
  - At Cornell University, “a **lakesource cooling system via heat exchange** with water from the bottom of nearby lake”.
  - In Herleen, NL, a **flooded mine** is used as an **open-loop thermal storage**
- **How can standardized methods or frameworks be developed for more broad deployment of DES?**
  - “Future District Heating Systems and Technologies: On the Role of Smart Energy Systems and 4th Generation District Heating.”
  - Examines systemic planning frameworks needed to successfully implement fourth-generation district heating.

Simpson, Juliet G., et al. “Decarbonized District Energy Systems: Past Review and Future Projections.” Science Direct, Elsevier, 24 Sept. 2024, www.sciencedirect.com/science/article/pii/S2590174524002046.
Lund, Henrick, et al. “Future District Heating Systems and Technologies: On the Role of Smart Energy Systems and 4th Generation District Heating.” Science Direct, Pergamon, 17 Sept. 2018, www.sciencedirect.com/science/article/abs/pii/S0360544218318796.

5 Unanswered Questions

- While this review includes a qualitative review of different types of DES, it lacks quantitative comparison.
- **How do different classes of DES compare under different climatic and demand profiles in quantitative terms? (e.g., emissions reduction potential, cost per kWh, etc.)**
  - "Assessment of district heating and cooling systems transition with respect to future changes in demand profiles and renewable energy supplies" by Yang, E. et al.
  - Presents information relevant to the cost effectiveness of systems based on different communities and the energy sources they have available to them.
- **How does DES compare quantitatively to alternative HVAC systems (such as individual standard AC and furnace split systems) servicing a community instead?**

Simpson, Juliet G., et al. "Decarbonized District Energy Systems: Past Review and Future Projections." Science Direct, Elsevier, 24 Sept. 2024, www.sciencedirect.com/science/article/pii/S2590174524002046.
Zhang, Yichi, et al. "Assessment of District Heating and Cooling Systems Transition with Respect to Future Changes in Demand Profiles and Renewable Energy Supplies." Science Direct, Pergamon, 31 July 2022, www.sciencedirect.com/science/article/pii/S0196890422008287.

5 Possible Extensions

- **More research** needs to be conducted on **decarbonized DES** in the US
  - **Few decarbonized installations** in the US
  - Current installations on college campuses or densely populated urban areas → **research new installations** in **other types of communities**
  - "Fifth-Generation District Heating and Cooling Networks Based on Shallow Geothermal Energy: A review and Possible Solutions for Mediterranean Europe" by Jordi García-Céspedes
  - The paper finds shallow geothermal energy to be well suited to the Mediterranean and standing as a top solution for decarbonization efforts

Simpson, Juliet G., et al. "Decarbonized District Energy Systems: Past Review and Future Projections." Science Direct, Elsevier, 24 Sept. 2024, www.sciencedirect.com/science/article/pii/S2590174524002046.

García-Céspedes, Jordi, et al. "Fifth-Generation District Heating and Cooling Networks Based on Shallow Geothermal Energy: A Review and Possible Solutions for Mediterranean Europe." MDPI, Multidisciplinary Digital Publishing Institute, 23 Dec. 2022, www.mdpi.com/1996-1073/16/1/147.

5 Possible Extensions

- Development of **Digital Twin Models for DES**
  - Develop digital twin models to "**answer difficult design and operational questions**". Modeling is a valuable **tool for assessment** of **designs** and **component sizing** without using any **expensive demonstration projects**.
- Research on **Improving currently installed DES**:
  - Focus on best practices and technical pathways to **retrofit older district systems** to **support low-carbon goals**.
  - Ball State University Examp(3G): large **borehole fields** combined with **heat recovery chillers replaced** the **original boilers and chillers**.

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5 Possible Extensions

UC San Diego

What are some possible extensions of DES technologies that you can think of beyond what's been listed?

Thank you!

Any questions?

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Extra figures

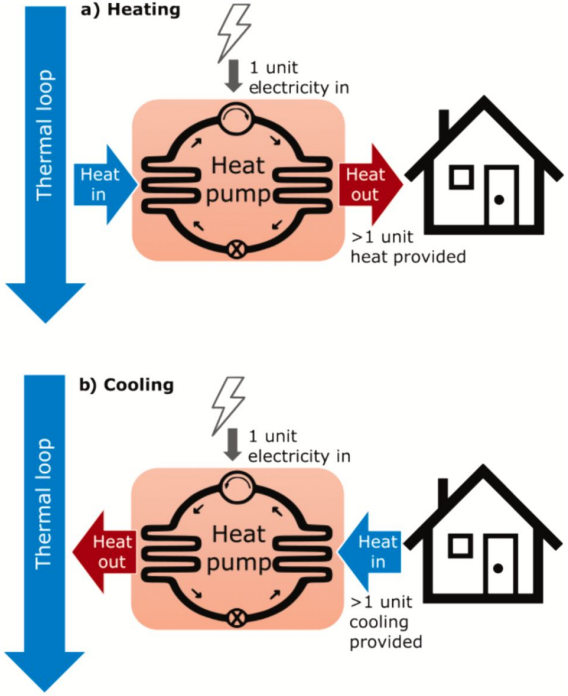

Fig. 4. Heat pump diagram in a) heating mode, and b) cooling mode.

Table 3
Summary of sources for decarbonized district heating and cooling.

Heat/Cool Source	Timing	Energy	Citations
CHP (nuclear, biomass, etc.)	Dispatchable	Electrical and high-temp	[15,36,37]
Shallow, vertical, or horizontal geothermal heat exchanger	Constant	Low-temp or heat storage	[38]
Deep geothermal	Constant	High-temp	[21,39]
Solar thermal	Variable	Medium-temp	[32]
Photovoltaic-thermal	Variable	Electrical and thermal	[40–42]
Natural thermal reservoirs (lake, river, ocean, etc.)	Constant	Low-temp or cooling	[33,43,44]
Waste heat	Variable	Medium or low-temp	[45–49,120]
Heat-powered cooling (absorption chiller)	Dispatchable	Cooling	
Electric-powered sources (heat pump, chiller)	Dispatchable	Heating and cooling	

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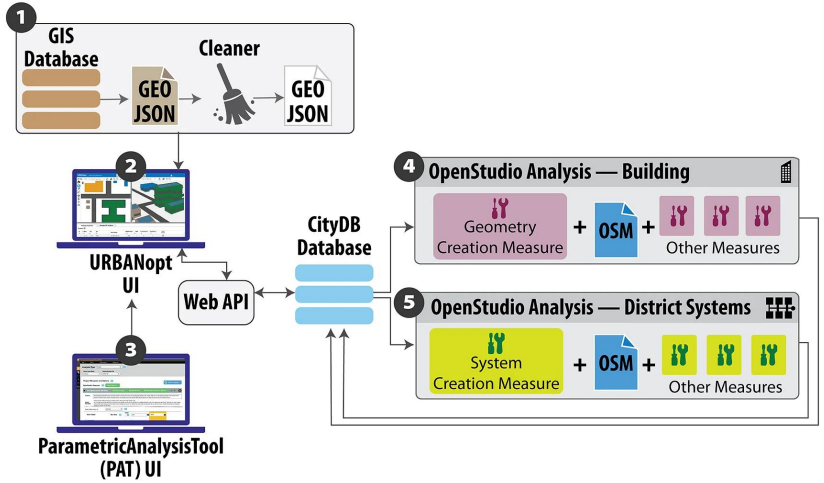

Figure: URBANopt platform architecture

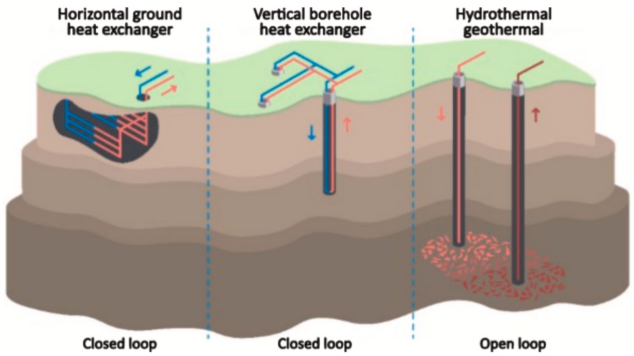

Fig. 5. Common types of geothermal systems used for district heating and cooling: shallow horizontal or vertical heat exchanger and deep hydrothermal. Adapted from [38].

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Appendix

UC San Diego

Acronyms

1G	first generation
2G	second generation
3G	third generation
4G	fourth generation
5G	fifth generation
AC	air conditioning
CHP	combined heat and power
COP	coefficient of performance
DES	district energy system
DHC	district heating and cooling
ETS	energy transfer station
GHG	greenhouse gas
HP	heat pump
LCOE	levelized cost of energy
NREL	National Renewable Energy Laboratory
PV	photovoltaic
RE	renewable energy
TEN	thermal energy network
UBEM	urban building energy modeling

Table 1. Selection of DES, mainly in North America, with novel decarbonized system aspects. Includes year installed (or upgraded), system type, storage, and thermal sources (with a focus on unique or decarbonized sources).

Site	Installed (or upgraded) Year	System Type	Thermal Source or Storage	Thermal Conversion	References
Colorado Mesa University. Colorado, USA	2008	5G DES, one pipe loop, ambient temp	471 boreholes, 91–152 m deep. Cooling towers, irrigation HX	heat pumps at users	[8], [19], [22], [23]
Mjinwater. Herleen, NL	2013	5G DES, two pipes, bidirectional flow	flooded mine as open-loop thermal storage	heat pumps at or near users	[24], [25], [26]
Whisper Valley "GeoGrid." Texas, USA	2017	5G DES, ambient temperature	237 boreholes at 107 m deep distributed along loop. Cooling tower	heat pumps at users	[27], [28], [29]
Eversource networked geothermal pilot. Massachusetts, USA	2023	5G DES, one pipe, ambient temperature	88 boreholes at 170–215 m deep	heat pumps at users	[30], [31]
Drake Landing Solar Community. Alberta, CA	2007	4G DES, two pipes	144 boreholes at 35 m deep. 240 m^3 hot water storage. Distributed solar thermal collectors	heat exchangers at users	[32]
Cornell University. New York, USA	2000	3G DES	lake-source cooling, heating via traditional sources (CHP)	heat exchangers at users	[33]
Ball State University. Indiana, USA	2012	3G DES, three pipes	3,600 boreholes at 122 m deep. Centralized heat recovery chillers	heat exchangers at users	[9], [34]

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